



SVP-18-075

10 CFR 50.73

November 6, 2018

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-29
NRC Docket No. 50-254

Subject: Licensee Event Report 254/2018-003-01 "Two Reactor Protection System Channels Impacted by Single Main Stop Valve Limit Switch Failure"

Enclosed is Licensee Event Report (LER) 254/2018-003-01, "Two Reactor Protection System Channels Impacted by Single Main Stop Valve Limit Switch Failure", for Quad Cities Nuclear Power Station, Unit 1.

This report is submitted in accordance with 10 CFR 50.73(a)(2)(vii) for an event where a single cause or condition caused two independent trains or channels to become inoperable in a single system designed to mitigate the consequences of an accident. The revision confirms the cause mechanism and the associated corrective actions.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Mark Humphrey at (309) 227-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "K. Ohr", written over a horizontal line.

Kenneth S. Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

TEZZ
NRR

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Quad Cities Nuclear Power Station Unit 1	2. Docket Number 05000254	3. Page 1 OF 3
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4. Title Two Reactor Protection System Channels Impacted by Single Main Stop Valve Limit Switch Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
06	09	2018	2018	003	01	11	06	2018	N/A	N/A

9. Operating Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level 100	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER	
Licensee Contact Richard Swart – Regulatory Assurance	Telephone Number (Include Area Code) (309) 227-2810

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
B	JD	33	N007	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No		N/A	N/A	N/A

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On June 9, 2018, at 1240 hours, Quad Cities Nuclear Power Station (QCNPS) main turbine Main Stop Valve number 1 (MSV1) failed to provide the appropriate Reactor Protection System (RPS) relay actuations during periodic surveillance activities. The associated limit switch failed to properly actuate relays in subchannels of both the A and B RPS Trip System channels. Operations entered Technical Specification (TS) 3.3.1.1. Condition B due to one inoperable channel in each trip system.

Fuses associated with the subchannels in both the A and B RPS channels were removed to meet the TS Action Statement. The limit switch was replaced on June 10, 2018, and associated RPS channels were tested satisfactory at 0538 hours. The cause of the switch failure was occasional binding due to inadequate lubrication during assembly by the manufacturer.

The safety significance of this event was minimal. The associated RPS logic requires signals from 3 of 4 valves, and the other 3 valves were providing proper signals per surveillance results. All other RPS inputs were functional. This report is submitted in accordance with the requirements of 10 CFR 50.73(a)(2)(vii) for a single condition that caused two independent channels to become inoperable in a single system designed to shut down the reactor and maintain it in a safe shutdown condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities Nuclear Power Station	05000254	2018	- 003	- 01

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Emergency Diesel Generators [DG]

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Two Reactor Protection System Channels Impacted by Single Main Stop Valve Limit Switch Failure

A. CONDITION PRIOR TO EVENT

Unit: 1 Event Date: June 9, 2018 Event Time: 1240 hours
Reactor Mode: 1 Mode Name: Power Operation Power Level: 100%

There were no structures, systems, or components (SSCs) out of service or inoperable that contributed to the event.

B. DESCRIPTION OF EVENT

On June 9, 2018, at 1240 hours, Quad Cities Nuclear Power Station (QCNPS) Unit 1 was performing a Turbine Stop Valve Closure Scram Instrumentation Functional Test. Main Stop Valve (MSV)[V] number 1 (MSV1), also referred to as Turbine Stop Valve number 1, did not produce the appropriate alarm or relay actuations in the Reactor Protection System (RPS)[JD] channels A or B. Operations entered Technical Specification (TS) 3.3.1.1. Condition B due to one inoperable channel in each trip system.

Operations subsequently removed fuses associated with the associated RPS sub-channels at 1838 on June 9, 2018 to satisfy the requirements of the TS Action Statement.

A new limit switch [33] was installed on the MSV number 1. Post maintenance testing on this new switch was satisfactory and TS 3.3.1.1. Condition B was exited at 0346 on June 10, 2018. The RPS logic for all four MSVs was verified satisfactory at 0538 on June 10, 2018.

This event is reportable as a licensee event report (LER) in accordance with 10 CFR 50.73(a)(2)(vii) for a single condition that caused two independent channels to become inoperable in a single system designed to shut down the reactor and maintain it in a safe shutdown condition.

C. CAUSE OF EVENT

An inspection of the degraded limit switch showed occasional binding attributed to insufficient or dried out lubricant, and a hardened o-ring on the operating shaft. Additional causal evaluation confirmed inadequate lubrication during assembly by the manufacturer.

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CONTINUATION SHEET**

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NARRATIVE**D. SAFETY ANALYSIS**

No actual safety consequences resulted from the failed limit switch since no related turbine generator trip requiring an RPS scram was experienced by QCNPS during this time. No automatic functions occurred as a result of the failed limit switch.

Although the limit switch for MSV1 would not have provided input to its associated RPS logic, the limit switches on the remaining three MSVs were verified to be properly functioning. Since any three MSVs can cause a scram from the RPS system logic, the scram would have occurred if required.

The MSV system design has all four valves closing on a turbine generator trip signal, supporting proper RPS function should it have been required. Individual MSV closures are only associated with surveillance testing.

A Probabilistic Risk Assessment (PRA) review indicated that the individual MSV limit switches are not explicitly represented in the QCNPS Full Power Internal Events PRA models. Due to system redundancy, combinations of multiple failures would be required for an RPS failure. Since component failure events were found to have failure probabilities less than 1E-06 per demand, such combinations of failure events have a negligible contribution to Core Damage Frequency risk.

This event is not considered a Safety System Functional Failure per NEI 99-02 Revision 7.

E. CORRECTIVE ACTIONS

Immediate Actions: The limit switch was replaced and tested with satisfactory results. RPS circuit fuses were replaced.

Follow Up Actions: A lubrication inspection will be invoked during the dedication process on this model of switch.

All other MSV limit switches on both units will be replaced at the next refuel outage and will be considered for replacement in a forced outage. The population of switches removed will be inspected for lubrication issues.

Potential Part 21 Notification initiated by Curtiss-Wright (October 8, 2018 letter to US NRC).

F. PREVIOUS OCCURRENCES

No previous similar events have occurred at the site based on a search of station LERs and IRs.

G. COMPONENT FAILURE DATA

Failed equipment: position indicating limit switch

Component manufacturer: NAMCO, with Commercial Grade Dedication by QualTech/Curtiss-Wright

Component model number: EA 700-90964

Component type: Snap-Lock cam operated switch

An ICES report will be submitted.